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EXAMPLES ON THE

And $\int \frac{dx}{\sqrt{2ax + x^2}} = \log \{x + a + \sqrt{2ax + x^2}\},$

$$\therefore \int \frac{x dx}{\sqrt{2ax + x^2}}$$

$$= \sqrt{2ax + x^2} - a \log \{a + a + \sqrt{2ax + x^2}\}.$$

$$\therefore \int \frac{x^2 dx}{\sqrt{2ax + x^2}} = \frac{x\sqrt{2ax + x^2}}{2} - \frac{3a}{2} \sqrt{2ax + x^2}$$

$$+ \frac{3a^2}{2} \log \{x + a + \sqrt{2ax + x^2}\}.$$

$$(3.) \int_a^o \frac{x^2 dx}{(2ax - x^2)^{\frac{3}{2}}}$$

let $p = x \quad dp = dx, \quad dq = \frac{x dx}{(2ax - x^2)^{\frac{3}{2}}},$

$$\therefore q = \int \frac{x dx}{(2ax - x^2)^{\frac{3}{2}}} = \int (2a - x)^{-\frac{3}{2}} x^{\frac{1}{2}} dx =$$

$$\int (2ax^{-1} - 1)^{-\frac{3}{2}} x^{-2} dx$$

$$= -\frac{1}{2a} \int (2ax^{-1} - 1)^{-\frac{3}{2}} \times -2ax^{-2} dx$$

$$= \frac{(2ax^{-1} - 1)^{-\frac{1}{2}}}{a} = \frac{1}{a} \sqrt{\frac{x}{2a - x}}.$$

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2 formula

And $\int \frac{dx}{\sqrt{2ax + x^2}} = \log \{x + a + \sqrt{2ax + x^2}\},$

3 formula

$$\therefore \int \frac{x dx}{\sqrt{2ax + x^2}}$$

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$$= \sqrt{2ax + x^2} - a \log \{a + a + \sqrt{2ax + x^2}\}.$$

5 formula

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6 formula

$$+ \frac{3a^2}{2} \log \{x + a + \sqrt{2ax + x^2}\}.$$

7 formula

$$(3.) \int_a^o \frac{x^2 dx}{(2ax - x^2)^{\frac{3}{2}}}$$

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9 formula

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10 formula

$$\int (2ax^{-1} - 1)^{-\frac{3}{2}} x^{-2} dx$$

11 formula

$$= -\frac{1}{2a} \int (2ax^{-1} - 1)^{-\frac{3}{2}} \times -2ax^{-2} dx$$

12 formula

$$= \frac{(2ax^{-1} - 1)^{-\frac{1}{2}}}{a} = \frac{1}{a} \sqrt{\frac{x}{2a - x}}.$$

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